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Science - Grade Four (#5020050)

Version: 2024 - And Beyond (current)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- [SC.4.E.5.1](#): Observe that the patterns of stars in the sky stay the same although they appear to shift across the sky nightly, and different stars can be seen in different seasons.
- [SC.4.E.5.2](#): Describe the changes in the observable shape of the moon over the course of about a month.
- [SC.4.E.5.3](#): Recognize that Earth revolves around the Sun in a year and rotates on its axis in a 24-hour day.
- [SC.4.E.5.4](#): Relate that the rotation of Earth (day and night) and apparent movements of the Sun, Moon, and stars are connected.
- [SC.4.E.5.5](#): Investigate and report the effects of space research and exploration on the economy and culture of Florida.
- [SC.4.E.6.1](#): Identify the three categories of rocks: igneous, (formed from molten rock); sedimentary (pieces of other rocks and fossilized organisms); and metamorphic (formed from heat and pressure).
- [SC.4.E.6.2](#): Identify the physical properties of common earth-forming minerals, including hardness, color, luster, cleavage, and streak color, and recognize the role of minerals in the formation of rocks.
- [SC.4.E.6.3](#): Recognize that humans need resources found on Earth and that these are either renewable or nonrenewable.
- [SC.4.E.6.4](#): Describe the basic differences between physical weathering (breaking down of rock by wind, water, ice, temperature change, and plants) and erosion (movement of rock by gravity, wind, water, and ice).

- [SC.4.E.6.5](#): Investigate how technology and tools help to extend the ability of humans to observe very small things and very large things.
- [SC.4.E.6.6](#): Identify resources available in Florida (water, phosphate, oil, limestone, silicon, wind, and solar energy).
- [SC.4.L.16.1](#): Identify processes of sexual reproduction in flowering plants, including pollination, fertilization (seed production), seed dispersal, and germination.
- [SC.4.L.16.2](#): Explain that although characteristics of plants and animals are inherited, some characteristics can be affected by the environment.
- [SC.4.L.16.3](#): Recognize that animal behaviors may be shaped by heredity and learning.
- [SC.4.L.16.4](#): Compare and contrast the major stages in the life cycles of Florida plants and animals, such as those that undergo incomplete and complete metamorphosis, and flowering and nonflowering seed-bearing plants.
- [SC.4.L.17.1](#): Compare the seasonal changes in Florida plants and animals to those in other regions of the country.
- [SC.4.L.17.2](#): Explain that animals, including humans, cannot make their own food and that when animals eat plants or other animals, the energy stored in the food source is passed to them.
- [SC.4.L.17.3](#): Trace the flow of energy from the Sun as it is transferred along the food chain through the producers to the consumers.
- [SC.4.L.17.4](#): Recognize ways plants and animals, including humans, can impact the environment.
- [SC.4.N.1.1](#): Raise questions about the natural world, use appropriate reference materials that support understanding to obtain information (identifying the source), conduct both individual and team investigations through free exploration and systematic investigations, and generate appropriate explanations based on those explorations.
- [SC.4.N.1.2](#): Compare the observations made by different groups using multiple tools and seek reasons to explain the differences across groups.
- [SC.4.N.1.3](#): Explain that science does not always follow a rigidly defined method ("the scientific method") but that science does involve the use of observations and empirical evidence.
- [SC.4.N.1.4](#): Attempt reasonable answers to scientific questions and cite evidence in support.

- [SC.4.N.1.5:](#) Compare the methods and results of investigations done by other classmates.
- [SC.4.N.1.6:](#) Keep records that describe observations made, carefully distinguishing actual observations from ideas and inferences about the observations.
- [SC.4.N.1.7:](#) Recognize and explain that scientists base their explanations on evidence.
- [SC.4.N.1.8:](#) Recognize that science involves creativity in designing experiments.
- [SC.4.N.2.1:](#) Explain that science focuses solely on the natural world.
- [SC.4.N.3.1:](#) Explain that models can be three dimensional, two dimensional, an explanation in your mind, or a computer model.
- [SC.4.P.8.1:](#) Measure and compare objects and materials based on their physical properties including: mass, shape, volume, color, hardness, texture, odor, taste, attraction to magnets.
- [SC.4.P.8.2:](#) Identify properties and common uses of water in each of its states.
- [SC.4.P.8.3:](#) Explore the Law of Conservation of Mass by demonstrating that the mass of a whole object is always the same as the sum of the masses of its parts.
- [SC.4.P.8.4:](#) Investigate and describe that magnets can attract magnetic materials and attract and repel other magnets.
- [SC.4.P.9.1:](#) Identify some familiar changes in materials that result in other materials with different characteristics, such as decaying animal or plant matter, burning, rusting, and cooking.
- [SC.4.P.10.1:](#) Observe and describe some basic forms of energy, including light, heat, sound, electrical, and the energy of motion.
- [SC.4.P.10.2:](#) Investigate and describe that energy has the ability to cause motion or create change.
- [SC.4.P.10.3:](#) Investigate and explain that sound is produced by vibrating objects and that pitch depends on how fast or slow the object vibrates.
- [SC.4.P.10.4:](#) Describe how moving water and air are sources of energy and can be used to move things.
- [SC.4.P.11.1:](#) Recognize that heat flows from a hot object to a cold object and that heat flow may cause materials to change temperature.
- [SC.4.P.11.2:](#) Identify common materials that conduct heat well or poorly.

- [SC.4.P.12.1](#): Recognize that an object in motion always changes its position and may change its direction.
- [SC.4.P.12.2](#): Investigate and describe that the speed of an object is determined by the distance it travels in a unit of time and that objects can move at different speeds.
- [MA.K12.MTR.1.1](#): **Actively participate in effortful learning both individually and collectively.**

Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.
- [MA.K12.MTR.2.1](#): **Demonstrate understanding by representing problems in multiple ways.**

Mathematicians who demonstrate understanding by representing problems in multiple ways:

- Build understanding through modeling and using manipulatives.
- Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.

- Progress from modeling problems with objects and drawings to using algorithms and equations.
- Express connections between concepts and representations.
- Choose a representation based on the given context or purpose.

Clarifications:

Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.
- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.

• **MA.K12.MTR.3.1: Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:

Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- **MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others.**

Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

- Communicate mathematical ideas, vocabulary and methods effectively.
- Analyze the mathematical thinking of others.
- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.
- Construct possible arguments based on evidence.

Clarifications:

Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

- **MA.K12.MTR.5.1: Use patterns and structure to help understand and connect mathematical concepts.**

Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

- Focus on relevant details within a problem.
- Create plans and procedures to logically order events, steps or ideas to solve problems.
- Decompose a complex problem into manageable parts.
- Relate previously learned concepts to new concepts.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.

Clarifications:

Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

- Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
- Support students to develop generalizations based on the similarities found among problems.
- Provide opportunities for students to create plans and procedures to solve problems.
- Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- **MA.K12.MTR.6.1: Assess the reasonableness of solutions.**

Mathematicians who assess the reasonableness of solutions:

- Estimate to discover possible solutions.
- Use benchmark quantities to determine if a solution makes sense.
- Check calculations when solving problems.
- Verify possible solutions by explaining the methods used.
- Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to assess the reasonableness of solutions:

- Have students estimate or predict solutions prior to solving.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"
- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justifications.

- **MA.K12.MTR.7.1: Apply mathematics to real-world contexts.**

Mathematicians who apply mathematics to real-world contexts:

- Connect mathematical concepts to everyday experiences.
- Use models and methods to understand, represent and solve problems.

- Perform investigations to gather data or determine if a method is appropriate.
- Redesign models and methods to improve accuracy or efficiency.

Clarifications:

Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
 - Challenge students to question the accuracy of their models and methods.
 - Support students as they validate conclusions by comparing them to the given situation.
 - Indicate how various concepts can be applied to other disciplines.
- [ELA.K12.EE.1.1](#): Cite evidence to explain and justify reasoning.

Clarifications:

K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- [ELA.K12.EE.2.1](#): Read and comprehend grade-level complex texts proficiently.

Clarifications:

See [Text Complexity](#) for grade-level complexity bands and a text complexity rubric.

- [ELA.K12.EE.3.1](#): Make inferences to support comprehension.

Clarifications:

Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like “Why is the girl smiling?” or make predictions about what will happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- [ELA.K12.EE.4.1](#): Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

Clarifications:

In kindergarten, students learn to listen to one another respectfully.

In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think _____ because _____.” The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- [ELA.K12.EE.5.1](#): Use the accepted rules governing a specific format to create quality work.

Clarifications:

Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- [ELA.K12.EE.6.1](#): Use appropriate voice and tone when speaking or writing.

Clarifications:

In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd

grade and beyond, students practice appropriate social and academic language to discuss texts.

- [ELD.K12.ELL.SC.1](#): English language learners communicate information, ideas and concepts necessary for academic success in the content area of Science.
- [ELD.K12.ELL.SI.1](#): English language learners communicate for social and instructional purposes within the school setting.
- [HE.4.C.1.5 \(Archived Standard\)](#): Identify the human body parts and organs that work together to form healthy body systems.

Course Information

General Notes

Special Notes:

Instructional Practices

Teaching from a range of complex text is optimized when teachers in all subject areas implement the following strategies on a routine basis:

1. Ensuring wide reading from complex text that varies in length.
2. Making close reading and rereading of texts central to lessons.
3. Emphasizing text-specific complex questions, and cognitively complex tasks, reinforce focus on the text and cultivate independence.
4. Emphasizing students supporting answers based upon evidence from the text.
5. Providing extensive research and writing opportunities (claims and evidence).

Science and Engineering Practices (NRC *Framework for K-12 Science Education*, 2010)

- Asking questions (for science) and defining problems (for engineering).
- Developing and using models.
- Planning and carrying out investigations.
- Analyzing and interpreting data.
- Using mathematics, information and computer technology, and computational thinking.